BAAJA: $50-100 billion in additional transmission payments

Transmission payments, 2027-2056 | October 31, 2026

What the bill would raise

We estimate that BAAJA would require data centers to pay an additional $56 billion for transmission, reducing other customers' bills by the same amount. It covers data centers added in 2027-2056, with charges starting in 2030. All figures are in 2026 dollars, without discounting. Selected cases yield $50-100 billion; other outcomes remain plausible. [1, 3]

The bill requires data centers to pay for their assigned upgrades and their share of the existing grid. Utilities must subtract the upgrade payments from the costs recovered through shared transmission rates. With the same investment and electricity demand, each extra dollar paid by data centers is a dollar other customers no longer owe. [1]

What drives the estimate

We use LBNL's projection of 649 TWh of data-center electricity use in 2030, then assume 3% annual growth through 2040 and 1.5% thereafter. We assume 80% load factor, 75% of new load covered by the bill, and $650,000/MW of assigned upgrades, rising 1% a year above inflation. Data centers already fund 35% of upgrade capital in the baseline. Capital is recovered over 40 years at 5% real financing, plus annual operating costs of 1.5% of capital. Existing-grid charges use PJM tariffs and a national starting estimate of $95,100/MW-year. Upgrade costs, coverage and existing funding are assumptions, not measured national averages. [2, 3, 4]

If data centers already fund 50% of upgrade capital, additional payments fall to $47 billion. Faster demand growth and $750,000/MW of upgrades raise them to $103 billion. These cases give the rounded $50-100 billion range. Other assumptions give lower results: covering only data centers added from 2030 yields $35 billion; using $300,000/MW of upgrades yields $26 billion. [3]

How it compares with the transmission tax credit

JCT estimated that the transmission investment tax credit in the House-passed 2021 Build Back Better bill would cost the federal government $11.279 billion over FY2022-2031. Converting each year to 2026 dollars at 2.5% inflation and repeating that ten-year schedule three times gives $32 billion over 30 years. This extension is our calculation; JCT did not score a 30-year program. [3, 5]

BAAJA's $56 billion central estimate is 1.8 times that amount. The $50-100 billion range is 1.6-3.2 times as large. This compares customer bill relief with the tax credit's federal budget cost, not the policies' economic benefits.

How much is above cost of service?

Using the larger of ordinary payments or incremental service cost as the benchmark, BAAJA payments are $55.6 billion higher. This result depends on how cost of service is defined. Counting a full share of the existing grid as well as assigned upgrades leaves no excess under this model. Payments exceed incremental cost alone by $145 billion, but $89 billion of that is already paid in the baseline. Only the additional payments measure what the bill changes. [3]

Sources

[1] S.5653, section 2107(a)(3)

[2] LBNL, U.S. Data Center Energy Usage, 2025 Update (June 2026)

[3] Review guide: bill provisions, source data, equations and calculation checks

[4] AEP, Piketon transmission investment and customer funding (March 2026)

[5] JCT, JCX-46-21 (November 19, 2021), p. 2, item F.1.5; p. 8, note 7

Methodology and full source links

Project load and calculate upgrade costs

We interpolate between LBNL's 192 TWh in 2024 and 464 TWh in 2028 to get 298.476 TWh in 2026, then reach 649 TWh in 2030. Growth after 2030 is our assumption. New covered MW = (annual TWh - 298.476) x 1,000,000 / (8,760 x 0.80) x 0.75. Each year's added MW costs $650,000/MW, rising 1% annually from 2026. [2, 3] The faster case uses 649 x EIA server-use[y]/server-use[2030], then extends the 2049-2050 growth rate through 2056. [6]

Annual capital recovery = capital x 0.05 / (1 - 1.05^(-40)). Add 1.5% of capital for operations and maintenance. In the baseline, data centers already pay 35% of capital recovery directly, but no operating costs directly. Existing load and network cost K grow 1% a year. Half of covered growth is assigned to PJM billing areas; the rest goes to the remaining U.S. system. These are model assumptions. [3, 4, 7-15]

Calculate who pays

For each region/year: K = annual existing-network cost; C = annual assigned-upgrade cost; D = annual equivalent of existing direct upgrade funding; s = new data centers' share of billing demand. The baseline collects D directly and shares the remaining costs. AND collects C directly and removes it from shared rates, leaving K. [1, 3]

Baseline = D + s(K + C - D). AND = C + sK. Extra payments = (1-s)(C-D).

Above higher-of = AND - max(baseline, C). Add results across regions and years 2030-2056.

Check the totals and compare with the ITC

In $bn: AND 232.975 - baseline 176.955 = 56.020 extra. AND - higher-of 177.396 = 55.579 above that benchmark. JCT's annual costs for 2022-2031 are 0, 0, 0, .788, 1.213, 1.213, 1.213, 2.001, 2.426, 2.425. Our 30-year ITC comparison = 3 x sum(cost[y] / 1.025^(y-2026)) = $31.674bn; 56.020 / 31.674 = 1.77. The extension is our calculation, not a JCT score; the ratio compares dollar amounts, not equivalent benefits. [3, 5]

Sources (same numbers as page 1, with additional inputs below)

[1] S.5653 IS: section 2107(a)(3)-(4), pp.229-231; separate FPA 228(f), p.240.
https://www.govinfo.gov/content/pkg/BILLS-119s5653is/pdf/BILLS-119s5653is.pdf

[2] LBNL 2025 Update (June 2026): printed pp.8-9, physical PDF pp.10-11.
https://www.energy.gov/documents/united-states-data-center-energy-usage-report-2025-update

[3] Review guide: input files, source cells, equations and calculation checks.
https://github.com/dashboardamerican/baaja-and-pricing/blob/main/analysis/fresh-central-2026-10-31/REVIEW_GUIDE.md

[4] AEP Piketon (March 2026): $4.2bn for 10GW; planned customer funding.
https://www.aep.com/news/stories/view/10823/

[5] JCT JCX-46-21: p.2, item F.1.5; p.8, note 7. Includes direct pay.
https://www.govinfo.gov/content/pkg/GOVPUB-Y4_T19_4-PURL-gpo172711/pdf/GOVPUB-Y4_T19_4-PURL-gpo172711.pdf

[6] EIA AEO2026 Table5: cb2026.d021826b, years C10:AB10; server energy C33:AB33.
https://www.eia.gov/outlooks/aeo/excel/aeotab5.xlsx

[7] PJM August2026 NITS: column E; exact zonal cells in [3].
https://www.pjm.com/-/media/DotCom/markets-ops/settlements/network-integration-trans-service-august-2026-xls.xlsx

[8] PJM 2026 network service peak loads: p.1, Zonal Peak (MW).
https://www.pjm.com/-/media/DotCom/markets-ops/settlements/network-service-peak-loads-2026.pdf

[9] PJM 2026 forecast: Table B9b, years B7:V7; zone rows8-31.
https://www.pjm.com/-/media/DotCom/planning/res-adeq/load-forecast/2026-load-report-tables.xlsx

[10] EIA AEO2026 Table54: C81, C136, D136, D140 (national expenditure proxy).
https://www.eia.gov/outlooks/aeo/supplement/excel/suptab_54.xlsx

[11] Grid Strategies 2025 load-growth report: p.33, 829GW planning-peak proxy.
https://gridstrategiesllc.com/wp-content/uploads/Grid-Strategies-National-Load-Growth-Report-2025.pdf

[12] Dominion October2024 local plan: selected needs/project pages indexed in [3].
https://www.pjm.com/-/media/DotCom/committees-groups/committees/teac/2024/20241008/20241008-informational-only---dom-local-plan-submission-of-supplemental-projects-2024.pdf

[13] MISO MTEP25: Appendix A New Local Reliability Projects, IDs50208 and50628.
https://cdn.misoenergy.org/MTEP25731615.zip

[14] MISO MTEP26 draft: upgrade examples; project matches and limitations in [3].
https://cdn.misoenergy.org/MTEP26%20-%20Draft784746.zip

[15] Entergy Kline testimony: printed pp.33-34 (PDF pp.35-36), customer funding and term.
https://lpscpubvalence.lpsc.louisiana.gov/portal/PSC/ViewFile?fileId=AUl4qI5mSHA%3D